

Lead Personalization System & Workflow Guide

Sequential Multi-Agent Pipeline utilizing FastAPI and Python

An in-depth guide to the 7-stage sequential AI pipeline engineered to automate healthcare prospect discovery, relevance scoring, signal-based research, and compliant outreach drafting without direct LinkedIn automation, scraping, or clinical data ingestion.

Document Scope:	Technical Blueprint and Operations Manual
Core Engine SDK:	FastAPI & Asynchronous Python
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1. Executive Summary & Core Objectives

VoiceCare AI's Go-To-Market (GTM) team requires a reliable, compliant, and cost-effective solution to surface high-value revenue cycle and patient access leaders. The system must research prospects using only public information, match recent professional events (signals) to VoiceCare AI use cases, and draft highly personalized messaging. Crucially, the system operates on a 'draft-only' paradigm where humans review, edit, and send outreach manually.

Core Goals and Fit Criteria:

- **Daily Pipeline Output:** Surface exactly 5 verified high-value prospects every run (no fillers).
- **Verifiable Signals:** Find at least one dated, public signal per prospect (job change, hiring, news).
- **VoiceCare Use-Cases:** Map prospect pain hypotheses directly to approved RCM and patient access use-cases.
- **Message Drafting:** Generate three ready-to-send outreach formats: connection note, follow-up, and email.
- **Zero Automation:** Drafting only. The system never connects, messages, or interacts on behalf of reps.
- **Free-Tier Stack:** Uses JINA APIs for search and scraping, and local file storage to operate at zero cost.

2. The 7-Stage Personalization Pipeline

The system utilizes a sequential multi-stage architecture to segment different cognitive tasks into sequential agent actions, passing validated context downstream.

- **Stage 01: Discovery:** Queries s.jina.ai for target roles (RCM Directors, Billing Mgrs) and ingests manual inputs.
- **Deduplication:** Checks the name, company, and LinkedIn URL against db.json (lookback default 45 days) and drops duplicates.
- **Stage 02: Scoring:** Evaluates fitting criteria out of 100. Selects top 5 and maps backfills.
- **Stage 03: Research:** Crawls company websites and news using Jina Reader to build an operational complexity profile.
- **Stage 04: Signal:** Extracts the single strongest dated signal. Low confidence scores trigger automated backfills.
- **Stage 05: Use-Case:** Maps signal and pain hypothesis to approved VoiceCare AI capabilities.
- **Stage 06: Drafting:** Generates connection invite notes, follow-up messages, and emails with no generic AI phrasing.
- **Stage 07: Review:** Automated compliance pre-check. Rejects and retries drafts up to 2 times before manual review.

3. Technical Architecture & Decisions

To satisfy cost and local deployment constraints, we structured the architecture to run fully in Python with zero third-party cloud database costs.

Major Architectural Components:

- **Backend Framework:** FastAPI serves as the asynchronous backend, facilitating easy local running and Vercel routing.
- **Agent Framework:** Asynchronous Python logic drives the model completions, using Pydantic response schemas for deterministic structured output.
- **Web Search Layer:** Jina Search (s.jina.ai) and Reader (r.jina.ai) handle public crawling for free without SerpAPI keys.
- **Storage Layer:** A local db.json file acts as the database for runs, history logs, and deduplication indexes, avoiding Supabase costs.

System Constraints and Caveats:

- **Context Lengths:** Jina crawls are capped at 5,000 to 10,000 characters to prevent model token limits and high execution latency.
- **Vercel Limits:** Pipeline steps run in parallel. Local execution has unlimited timeouts, while Vercel routes are configured for 60 seconds.
- **Idempotency:** Re-running a date overwrites prior pipeline outputs for that date but preserves manual human approval decisions.

4. Dynamic Scheduler & Resend Integration

To automate lead personalization campaigns, we implemented a serverless-ready scheduler that operates autonomously via cron execution and emails HTML reports.

Scheduler Components & Logic:

- **Vercel Cron Trigger:** An hourly cron triggers `/api/cron-run` on Vercel Pro, or once-daily at 09:00 UTC (`0 9 * * *`) on Vercel Hobby, maintaining compatibility across all plans.
- **Dynamic Interval Filtering:** Before executing, the backend checks database settings in `db.json` to enforce frequency intervals: Daily (22h lookback), Alternate Days (46h lookback), or Weekly (166h lookback) since the last execution.
- **Resend HTML Report Dispatch:** If new prospects are surfaced, the scheduler formats and dispatches an executive daily report to the recipient email (defaulting to `shivamsingh0013@gmail.com`).
- **Passcode Decoupling:** Decouples Vercel's auto-injected `CRON_SECRET` from manual UI actions by supporting a custom `WORKSPACE_PASSCODE` variable, allowing users to trigger runs without copy-pasting the cron secret.

5. Resilient Fallbacks & Interactive Canvas UI

The system has been hardened against API limits and optimized for an intuitive, non-technical, and highly interactive user experience.

System Integration and UI Details:

- **Search Engine Fallback Chain:** Queries are resolved in the order: Tavily Search (primary search) -> Firecrawl Search (fallback 1) -> Jina Search (fallback 2) if keys are missing or rate-limited.
- **Sales Navigator Leads API:** Queries LinkedIn Sales Navigator for prospects, using a high-fidelity mock dataset fallback if keys are missing, preventing pipeline halts.
- **Figma-like Canvas Navigation:** Implements Spacebar / Middle Click click-and-drag panning, trackpad pinch-to-zoom, and Ctrl/Cmd + Scroll wheel zoom centered directly on the mouse cursor position.
- **Actionable Empty State:** Output Console drawer is accessible on page load. If clicked before a run, it shows a clean tutorial layout and a direct CTA button to run the agent.

6. Compliance & Safety Boundaries

VoiceCare AI operates in the healthcare sector, which demands absolute adherence to privacy regulations (HIPAA) and platform anti-spam rules. The system implements a strict safety core.

Compliance Policies Enforced:

- **No Scraping:** We do not scrape or automate logged-in LinkedIn pages. Only public search snippets are indexed.
- **No Clinical Data:** The pipeline ignores, rejects, and never stores patient-level data or PHI (Protected Health Information).
- **Fact Grounding:** Every claim or signal used in an outreach draft must be tied to a traceable source URL in db.json.
- **Tone Compliance:** Clichés, fake familiarity, and overpromised ROI outcomes are checked and rejected during Stage 07.
- **Manual Control:** A human reviewer must explicitly tick the validation checklist before any draft can be exported or approved.

TECHNICAL COMPLIANCE STATEMENT

This agentic architecture is 100% compliant with LinkedIn's terms of service and HIPAA safety boundaries. It acts purely as a local research assistant and drafting tool.